

Nirma University
Institute of Technology
Semester End Examination (IR), January - 2022
Master of Computer Application, Semester - I
3MCA104 Database Management System

Roll /
Exam No.

Supervisor's initial
with date

Time: 2 Hours

Max. Marks: 50

- Instructions:
1. Attempt all questions.
 2. Figures to right indicate full marks.
 3. Draw neat sketches wherever necessary.
 4. Assume suitable data wherever necessary and mention the same.

Q-1. Answer the following. [18]

- (A) What do you understand by data model? Discuss the different categories of data models. (6)
CO-1, BL-2
- (B) Why do we need a SQL key in the database management system? Discuss super key, candidate key, composite key, and alternative key by taking an appropriate example. (6)
CO-2, BL-1
- (C) Discuss union, intersect, and except operations over two sets by taking the appropriate examples of SQL. (6)
CO-2, BL-4

OR

- (C) What do you understand by transaction? Discuss ACID properties in order to ensure accuracy, completeness, and data integrity. (6)
CO-2, BL-2

Q-2. Answer the following. [14]

- (A) Write a PL/SQL program to demonstrate a User-defined Exception. Assume and mention the necessary data if required. (7)
CO-4, BL-6
- (B) Why normalization process in a database management system is important? (7)
CO-2, BL-1
- Discuss first normal form(1NF), second normal form(2NF), and third normal form(3NF) with appropriate examples.

Q-3. Answer the following. [18]

- (A) Write SQL for creating a table corresponding to the schema as mentioned below: (9)
CO-3, BL-6
- department(dept name, building, budget)
course(course id, title, dept name, credits)
instructor(ID, name, dept name, salary)
section(course id, sec id, semester, year, building, room number, time slot id)
teaches(ID, course id, sec id, semester, year)

Make any reasonable assumptions about data types, primary key and reference key.

(B) What is the objective of creating a trigger? Write a PL/SQL program to demonstrate a trigger for inserting data in a table. Assume and mention the necessary data if required. (9)
CO-4, BL-6

OR

(B) Write the steps to implement an explicit cursor. Also, write a PL/SQL program to demonstrate an explicit cursor that fetches the data of selected attribute and display the same. Assume and mention the necessary data if required. (9)
CO-4, BL-2
